

Pneumonia-CVD Colocalization

Bioinformatics 1 - Project 3 | 28 slides | 45 min talk + 30 min discussion | Supervisor: Prof. Cherng Yang

Speaker	Role	Slides	Section
● Hamza Arbaoui	Interpretation & Writing	1 - 5	Introduction, hypothesis & study design
● Jaouad El Morabit	Data & QC	6 - 14	Concepts, data, QC, harmonization & GWAS scan
● Cheikh Taha	Colocalization	15 - 22	Coloc results: 4 loci, master result & liver
● Loua Celistin	Visualization	23 - 25	CVD gene overlap, Venn diagrams & biology
● Hamza Arbaoui	(closing)	26 - 28	Conclusion, limitations & acknowledgements

Timing guide: ~11 min per speaker. Handoffs at section boundaries. Loua's block is figure-dense (all figures were prepared by her).

The full 28-slide deck follows this page.

BIOINFORMATICS 1 - PROJECT 3

Shared Genetic Architecture between Bacterial Pneumonia and Cardiovascular Disease

A GWAS-eQTL colocalization analysis - Pneumonia pilot

Jaouad El Morabit (Data & QC) - C. Taha - L. Celistin - H. Arbaoui

Supervisor: Prof. Cherng Yang - 16 July 2026

What this talk covers

- | | | |
|---|--|---|
| 1 | Background & hypothesis | Why link infection and CVD |
| 2 | Concepts | GWAS, eQTL, colocalization (PP.H0-H4) |
| 3 | Workflow & data | FinnGen GWAS x GTEx eQTL, QC, harmonization |
| 4 | Colocalization results | Four pneumonia loci tested |
| 5 | CVD gene overlap | Candidate genes, Venn diagram |
| 6 | Interpretation & conclusion | Biology, limitations, next steps |

Two disease burdens, one possible genetic root

The observation

- Bacterial infections (pneumonia, sepsis...) and cardiovascular disease are leading causes of death worldwide.
- Epidemiology links them: infection raises short- and long-term cardiovascular risk.
- Both involve immune activation and inflammation - shared biology is plausible.

The question

- Do infection susceptibility and CVD share **CAUSAL** genetic variants?
- If a variant drives both, it likely acts through immune / inflammatory gene regulation.
- Testable with genetics: overlap between GWAS signals and expression QTLs (eQTL).

Central hypothesis and analytical logic

Hypothesis: bacterial-infection susceptibility and CVD share causal variants acting through immune and inflammatory pathways, detectable as overlap between GWAS signals and tissue-specific eQTLs.



Scope (supervisor's brief): build ONE complete, robust pipeline on pneumonia first; extend the same workflow to other traits only if it works. The deliverable is a presentation, not software.

Six infection traits, tissue-specific eQTLs

Every trait: whole blood + one trait-relevant tissue (fixed by the supervisor). Pneumonia is the pilot.

Infection trait	Tissue 1	Tissue 2
Pneumonia (pilot)	Whole blood	Lung
Infective endocarditis	Whole blood	Artery (tibial)
Sepsis	Whole blood	Liver
Periodontitis	Whole blood	Artery (tibial)
H. pylori	Whole blood	Stomach
Tuberculosis	Whole blood	Lung

Pilot tissue rationale: lung = primary infection site; whole blood = shared immune readout for all traits.

GWAS and eQTL - the two signals we compare

G GWAS

- Associates each SNP with a trait (here: pneumonia case/control).
- Output per SNP: effect size, standard error, p-value, allele frequency.
- **Tells us WHERE on the genome disease risk sits - not which gene.**

E eQTL

- Associates each SNP with the expression of a nearby gene, in a given tissue.
- Source: GTEx v8 (whole blood, lung).
- **Tells us which gene a variant REGULATES, and in which tissue.**

Colocalization: five competing hypotheses

For a region, coloc.abf returns posterior probabilities PP.H0-PP.H4 that sum to 1:

H0

No association with either trait

H1

Association with GWAS only

H2

Association with eQTL only

H3

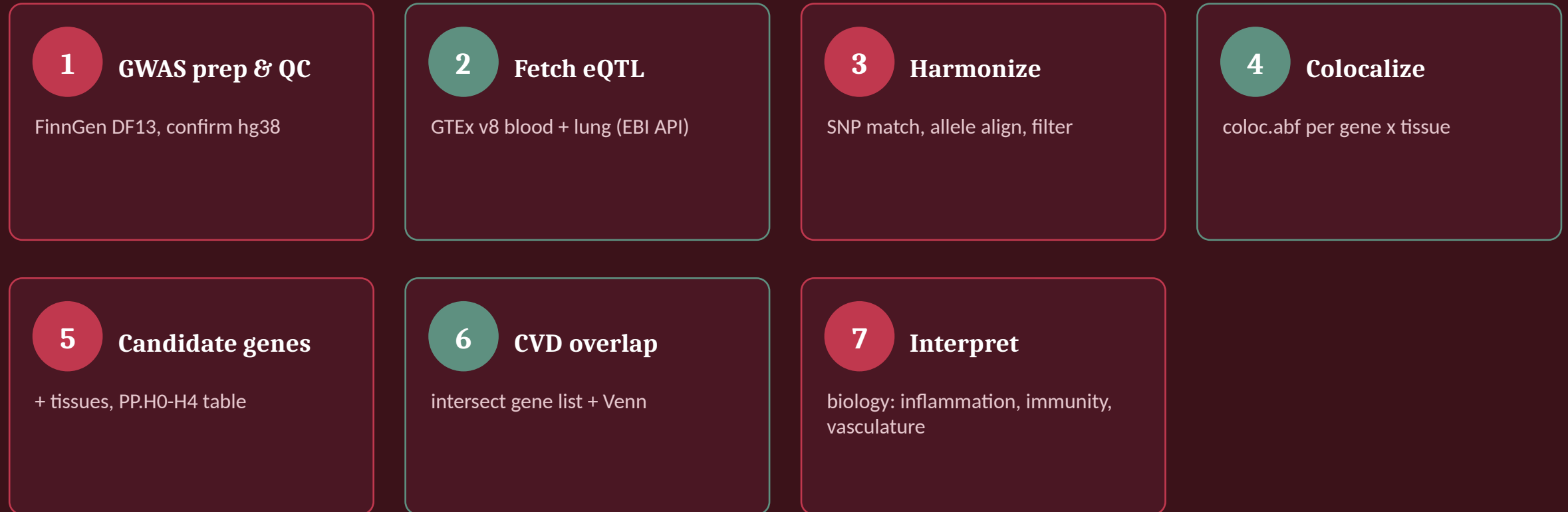
Both associated - **DIFFERENT** causal variants

H4

Both associated - **SAME** (shared) causal variant

We keep a colocalization when $PP.H4 > 0.5$. H4 = shared cause (interesting); H3 = same region, different variants (not shared).

Prof. Yang's pipeline, applied end-to-end



Threshold set by supervisor: $PP.H4 > 0.5$

Three inputs - and why

Input	Source	Why
Pneumonia GWAS	FinnGen Data Freeze 13 (J10_PNEUMONIA)	European ancestry (matches GTEx); hg38; 21.3M variants
eQTL (blood, lung, liver)	EBI eQTL Catalogue v2 (GTEx v8)	Region-queryable API; liver added for CRP/HNF1A (D3)
CVD gene list	cvd_genes.txt (supervisor)	4,309 HGNC symbols for gene-level overlap

Coloc parameters (locked): N = 500,186 | cases = 84,364 | s = 0.1687 | case-control. Both datasets hg38 -> no liftOver.

Genome build - the #1 failure point

The risk

- GWAS and eQTL must be on the SAME genome build.
- Mixing hg19 and hg38 shifts positions by thousands of bp - SNPs silently mis-match.
- coloc then compares the wrong variants.

Our check

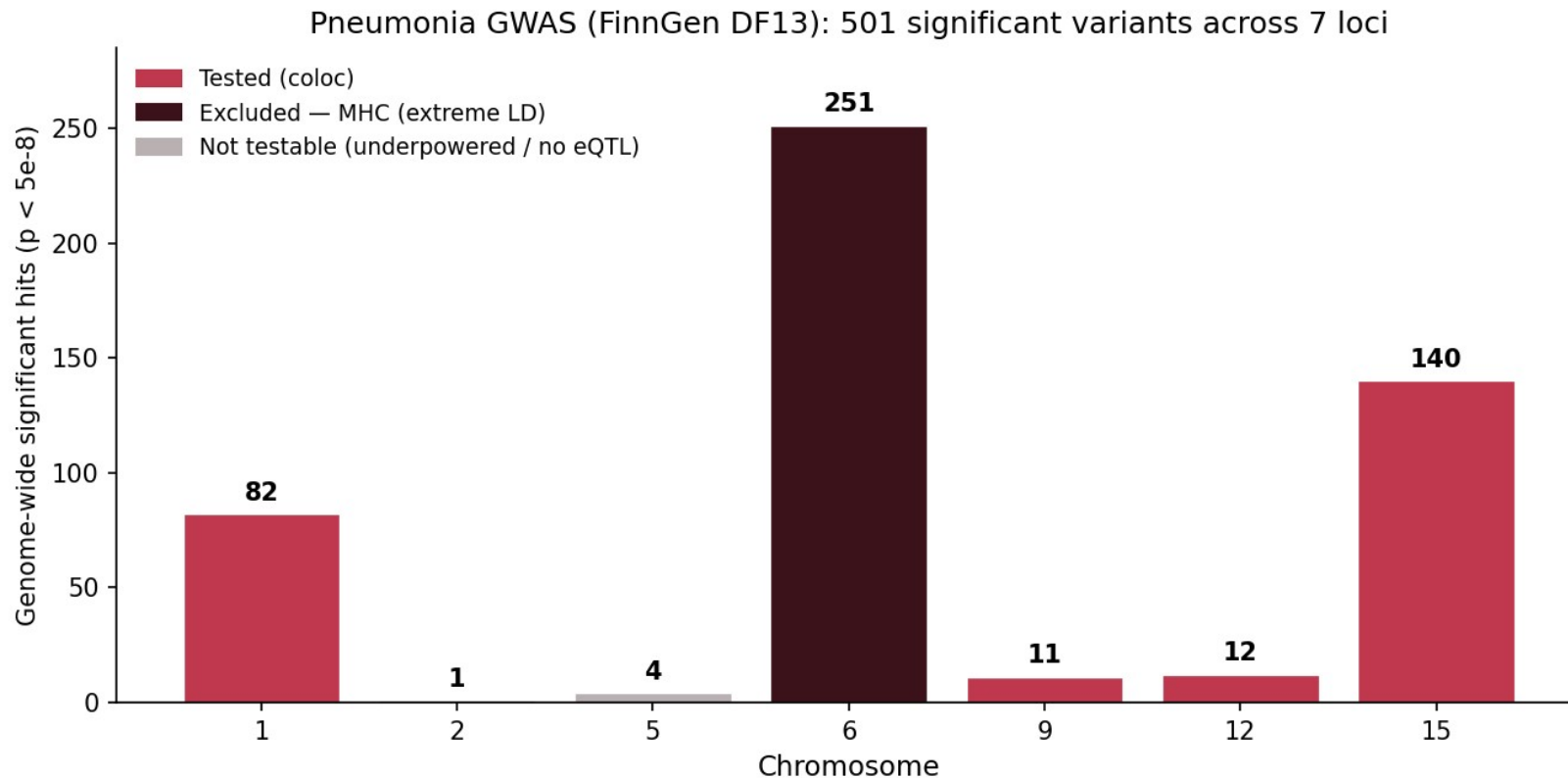
- **FinnGen DF13 = hg38. GTEx v8 = hg38.**
- Same build -> no liftOver needed (verified on a shared lead SNP).
- MAF from allele frequency; ambiguous strand SNPs handled at harmonization.

Harmonization - three checks per SNP

- 1 Match SNPs** by chromosome + position + sorted allele pair, per gene region
- 2 Align effect allele** flip the eQTL beta sign when ref/alt are swapped vs the GWAS
- 3 Drop ambiguous SNPs** remove A/T and C/G variants near MAF 0.5 (strand-flip risk)

Kept per region: only SNPs present in BOTH datasets. Late fix: coloc.abf needs unique SNPs -> duplicate multiallelic keys removed first.

GWAS scan: where are the signals?



Read-out

- 501 genome-wide significant variants.
- 7 loci across the genome.
- chr6 (MHC) largest but excluded.
- 4 clean loci taken forward.

Does the pipeline work? Recover known genes

Published pneumonia GWAS report...

- UK Biobank + FinnGen meta-analysis: two chr15 loci, HYKK and smoking genes CHRNA3 / CHRNA5.
- Nature Communications 2022: MUC5AC, TNFRSF1A (fine-mapped).

...our pipeline recovers

- **chr15 lead: HYKK, CHRNA3, CHRNA5 (recovered)**
- Same 15q25 locus as the published studies.
- MUC5AC / TNFRSF1A not recovered - different study, larger N, nearest-gene limits.

Recovering HYKK + CHRNA3/5 confirms the pipeline reproduces established biology.

Loci taken to colocalization

Locus	Hits	Decision
chr15 (15q25)	140	Tested - blood + lung
chr1 (CRP)	82	Tested - blood + lung
chr12 (HNF1A)	12	Tested - blood + lung
chr9 (TNFSF15)	11	Tested - blood + lung
chr2 (IFIH1)	1	Not testable - no IFIH1 eQTL
chr5 (CDH6)	4	Skipped - underpowered
chr6 (MHC)	251	Excluded - extreme LD

Selection rule

- Enough GWAS signal to define a peak.
- A cis-eQTL available in blood/lung.
- Not in the MHC.
- **4 loci qualify.**

15q25: the smoking / immune locus

Candidate genes (both signals present)

- CTSH (cathepsin H) - candidate #1, both tissues, immune gene
- IREB2, CIB2, TBC1D2B (blood); RASGRF1 (lung)
- 140 GWAS hits; strong eQTLs alongside

Colocalization outcome

- PP.H3 approx 1 for every strong gene
- PP.H4 max approx 0.02 - no colocalization
- Two real signals, different causal variants

15q25 is a known MULTI-signal (smoking) locus - a first hint that single-variant coloc may be limited here.

CRP: the inflammation locus

Candidate genes (both signals present)

- SLAMF8, IGSF8, COPA, NCSTN, AIM2, MNDA
- Region dense with immune / inflammatory genes
- CRP gene itself has no blood/lung eQTL (liver gene)

Colocalization outcome

- PP.H3 approx 1 across strong genes
- PP.H4 max 0.10 (H1-driven, not shared)
- No colocalization

Biologically rich region for the hypothesis - but still distinct causal variants.

HNF1A: an inflammation-to-CVD bridge

Candidate genes (both signals present)

- P2RX4 - purinergic receptor, inflammation & endothelial function
- COQ5, C12orf43, KDM2B
- HNF1A regulates CRP; no HNF1A eQTL in blood/lung

Colocalization outcome

- PP.H3 approx 1 (incl. P2RX4)
- PP.H4 max 0.29 - highest of the study, still < 0.5
- No colocalization

Best PP.H4 anywhere (0.29) is a non-coding RNA - not a shared-cause signal.

TNFSF15: testing the hypothesis gene directly

Candidate genes (both signals present)

- TNFSF15 itself HAS a strong blood eQTL - ideal test
- ORM1 (orosomucoid, acute-phase protein)
- Lung underpowered here (GWAS p only $\sim 1e-3$)

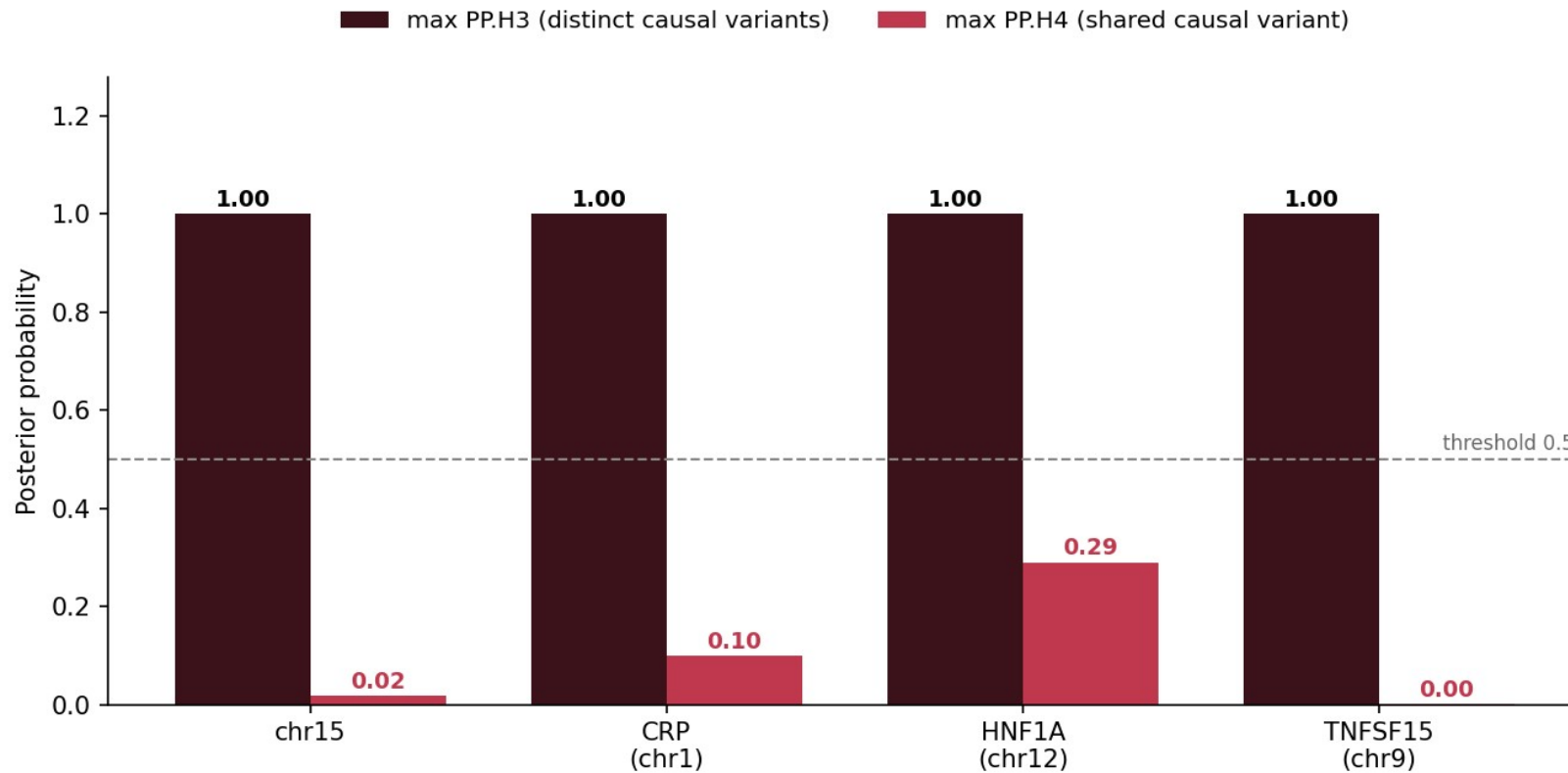
Colocalization outcome

- TNFSF15: PP.H3 = 0.999 (blood)
- Even the ideal gene -> distinct causal variants
- No colocalization

The clearest possible on-hypothesis test still returns H3, not H4.

RESULTS (8)

The result, in one figure



Master result

- 74 gene x tissue tests (3 tissues).
- **0 colocalizations (no PP.H4 > 0.5).**
- PP.H3 approx 1 wherever both signals are strong.

Why H3 everywhere? Two readings

Reading A - biological

- Pneumonia risk at these loci is NOT mediated by cis-expression of these genes in blood/lung.
- Real, interpretable result - the signals simply are not the same variant.

Reading B - methodological

- coloc.abf assumes ONE causal variant per region.
- chr15, CRP, HNF1A are multi-signal -> H3 can mask a hidden colocalization.
- **coloc.susie handles multiple signals (needs an LD panel) - the rigorous next step.**

Honest limits: loci we could not test

chr2 - IFIH1

Only 1 GWAS hit AND no cis-eQTL for IFIH1 in blood/lung. Colocalization is impossible - the data does not exist in these tissues.

chr5 - CDH6

Too few hits (4) - underpowered to define a peak for coloc.

chr6 - MHC

Excluded by design: extreme, extended LD makes colocalization uninterpretable across immune traits.

Does liver rescue CRP / HNF1A?

Why liver?

- CRP and HNF1A are primarily LIVER-expressed and had no blood/lung eQTL.
- Liver is the natural tissue to test them - added on the supervisor's suggestion (D3).
- GTEx v8 liver eQTL (EBI Catalogue) fetched for the CRP and HNF1A loci.

What we found

- **CRP and HNF1A still have NO cis-eQTL in liver -> lead genes remain untestable.**
- Where a neighbouring gene was testable -> PP.H3 approx 1 again.
- **No colocalization in liver either.**

Take-away: the result holds across THREE tissues (blood, lung, liver). The limitation is confirmed, not resolved - a robust, honest finding.

Candidate genes and tissues

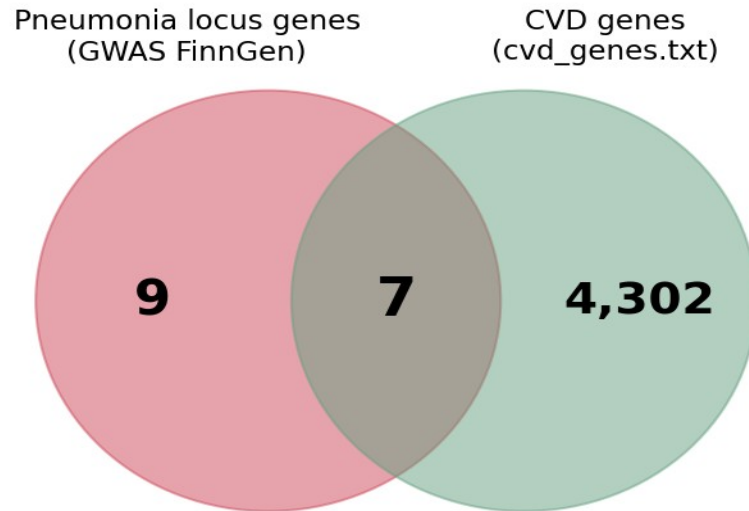
26 candidate genes (a GWAS AND an eQTL signal) across the four loci. * = also in the CVD list.

Locus	Candidate genes (double signal)	Tissue
chr15	CTSH*, IREB2, CIB2, TBC1D2B, RASGRF1	blood + lung
chr1 (CRP)	IGSF8*, PIGM*, CASQ1*, SLAMF8, COPA, NCSTN	blood + lung
chr12 (HNF1A)	P2RX4*, COQ5*, C12orf43, POP5, KDM2B	blood + lung
chr9 (TNFSF15)	TNFSF15, ORM1*, WHRN	blood

* gene is also in the CVD list (cvd_genes.txt).

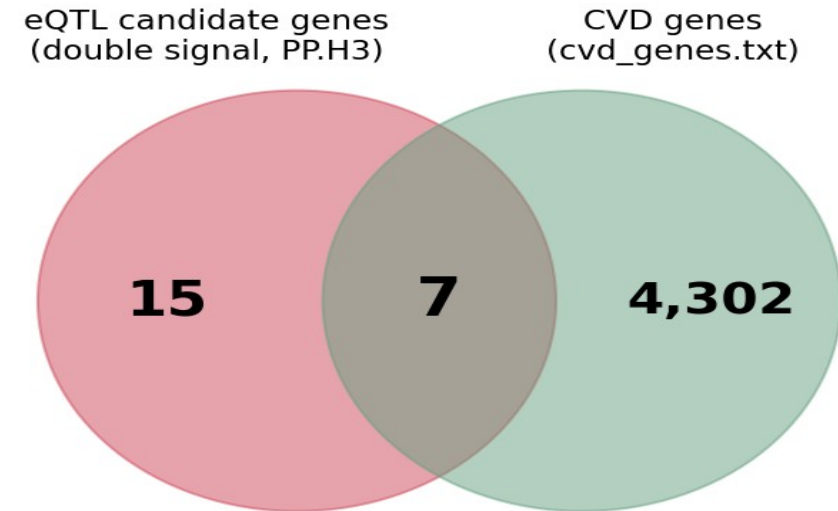
Gene-level overlap with CVD

Pneumonia GWAS genes vs CVD genes - 7 / 16 shared



Shared (7): *CHRNA3, CHRNA5, CRP, HCP5, HLA-DQB1, HNF1A, PSMA4*

eQTL candidate genes vs CVD genes - 7 shared



Shared (7): *CASQ1, COQ5, CTSH, IGSF8, ORM1, P2RX4, PIGM*

Caveat (stated honestly): the CVD list has 4,309 genes (~20% of protein-coding genes) - some overlap is expected by chance. A hit is suggestive, not proof; strengthened only by high PP.H4 (we have none) or clear biology.

Biology of the shared genes

P2RX4

Purinergic receptor - inflammation (NLRP3) & endothelial / vascular function

CTSH

Cathepsin H - antigen processing, immunity; present in both tissues

ORM1

Orosomucoid - acute-phase inflammation protein

CRP / HNF1A

Direct inflammation-to-CVD links (CRP level, cardiometabolic regulator)

These sit on the inflammation / immunity axis of the hypothesis - but colocalization did not confirm a shared causal variant.

CONCLUSION

Shared genes, but not shared causal variants

- Pneumonia and CVD SHARE loci and genes: 7/16 GWAS-locus genes and 7 candidate genes are CVD genes (incl. CRP, HNF1A, P2RX4).
- But colocalization finds NO shared causal variants: PP.H3 approx 1 across all four loci.
- The overlap is at the gene / region level, not the causal-variant level - under coloc.abf's single-variant assumption, in blood, lung and liver.
- **A consistent, reproducible result - and a validated, reusable pipeline for the other five traits.**

What would strengthen this

Limitations

- Single-causal-variant assumption at multi-signal loci.
- Three tissues tested (blood, lung, liver) - lead genes still lack cis-eQTL.
- No cis-eQTL for lead genes CRP, IFIH1, HNF1A.
- Broad CVD list -> chance overlap.

Perspectives

- coloc.susie on multi-signal loci (with 1000G LD).
- Add disease-relevant tissues (liver, artery).
- Report PP.H4 at 0.5 and 0.8 (sensitivity).
- eQTLGen (~31k) as a whole-blood supplement.
- Extend the pipeline to the other 5 traits.

TEAM & ACKNOWLEDGEMENTS

Thank you

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Interpretation & Writing

Supervisor: Prof. Cherng Yang - Pipeline based on his Project 3 workflow.

Data: FinnGen DF13 - GTEx v8 via EBI eQTL Catalogue - coloc (R). Validation: Nat Commun 2022; UKB+FinnGen meta-analysis.

Questions & discussion